

LEARMONTH, Australia

WMO: 943020

Lat: 22.2406S

Lon: 114.0967E

Elev: 6

StdP: 101.26

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.9	10.0	-6.1	2.3	26.8	-3.4	2.8	25.5	10.2	20.7	9.3	21.0	3.1	230	0.492

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	12.3	41.2	21.0	39.6	21.0	38.1	21.1	26.5	31.0	25.9	30.7	25.4	30.5	6.2	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.3	20.5	28.4	24.8	19.8	28.2	24.0	18.9	28.1	82.8	30.9	80.3	30.6	78.0	30.4	33.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.5	9.6	9.0	DB	5.9	45.2	1.1	1.4	5.1	46.2	4.5	47.1	3.9	47.9	3.1	48.9	(4)
(5)			WB	3.5	27.6	1.0	1.4	2.7	28.6	2.1	29.4	1.5	30.1	0.8	31.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	24.7	30.1	30.3	29.6	26.7	22.2	18.8	17.8	19.2	21.5	24.7	26.5	28.7	(6)	
	DBStd	4.99	2.34	2.21	2.21	2.22	2.41	2.28	2.00	2.05	2.06	2.26	2.38	2.55	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	71	0	0	0	0	1	22	33	14	1	0	0	0	(9)	
	CDD10.0	5349	624	569	608	502	379	263	243	284	343	456	495	581	(10)	
	CDD18.3	2378	366	336	349	252	122	35	18	40	94	198	245	323	(11)	
	CDH23.3	26952	4516	4247	4334	2723	1031	194	136	418	954	2042	2609	3749	(12)	
(13)	CDH26.7	14113	2638	2400	2387	1309	307	13	11	76	322	1030	1425	2197	(13)	
(14)	Wind	WSAvg	4.9	6.1	5.5	5.0	4.2	3.6	3.6	4.0	4.9	5.7	5.9	6.1	(14)	
(15) Precipitation	PrecAvg	232	33	39	33	17	41	46	21	12	2	2	1	1	(15)	
	PrecMax	615	361	239	305	130	231	224	99	64	15	46	32	11	(16)	
	PrecMin	37	0	0	0	0	0	0	0	0	0	0	0	0	(17)	
	PrecStd	134	71	58	62	29	48	46	25	17	4	7	5	3	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.0	42.9	41.2	38.5	34.0	28.1	28.0	31.0	35.0	38.4	40.1	43.1	(19)	
		MCWB	20.6	21.7	21.1	19.6	17.4	17.2	15.4	15.0	16.6	17.9	19.0	20.8	(20)	
	2%	DB	41.1	40.4	39.1	36.2	31.5	26.6	26.1	28.8	32.0	36.0	37.7	40.1	(21)	
		MCWB	21.2	22.0	21.4	19.3	17.7	17.1	15.7	15.2	15.9	17.9	18.9	19.9	(22)	
	5%	DB	38.9	38.5	37.4	34.5	29.8	25.5	24.9	27.1	30.0	34.1	35.9	38.1	(23)	
		MCWB	21.8	22.4	21.8	19.6	18.1	16.9	15.4	15.3	16.0	17.5	18.8	20.4	(24)	
	10%	DB	36.8	36.6	35.7	32.9	28.2	24.3	23.7	25.8	28.2	32.2	33.9	36.0	(25)	
		MCWB	22.3	22.8	22.1	20.1	18.4	16.8	15.3	15.2	16.0	17.3	18.9	20.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.1	27.2	27.1	25.6	24.1	22.6	21.5	20.8	21.0	22.3	23.5	26.1	(27)	
		MCDB	31.3	31.7	30.9	28.6	26.5	24.5	23.6	24.3	26.4	28.8	30.4	30.6	(28)	
	2%	WB	26.2	26.5	26.2	24.6	22.9	21.4	19.9	19.5	19.7	21.0	22.5	24.7	(29)	
		MCDB	30.9	31.2	30.6	28.3	25.8	23.4	22.1	23.1	24.9	27.9	29.7	31.5	(30)	
	5%	WB	25.6	25.9	25.6	23.7	21.9	20.2	18.8	18.3	18.9	20.0	21.5	23.7	(31)	
		MCDB	30.9	30.9	30.2	28.0	25.4	22.6	21.4	22.8	24.6	27.6	29.1	31.3	(32)	
	10%	WB	24.9	25.4	24.9	22.8	20.8	19.1	17.6	17.2	18.0	19.2	20.8	22.8	(33)	
		MCDB	31.1	30.6	29.9	28.0	24.7	22.2	20.8	22.4	24.5	26.9	28.2	31.1	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	13.6	12.3	12.4	12.3	12.1	11.4	12.2	13.9	14.8	15.7	15.0	14.8	(35)	
		MCDBR	17.4	16.0	15.0	14.6	14.0	12.3	13.7	15.4	17.0	18.1	17.8	18.0	(36)	
	5% WB	MCWBR	4.0	4.0	4.1	5.0	6.1	5.8	6.1	6.4	5.7	4.6	4.2	4.0	(37)	
		MCWBR	10.9	9.9	9.9	10.5	9.7	7.9	9.3	12.7	13.9	15.3	14.4	13.3	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.386	0.389	0.370	0.349	0.325	0.318	0.312	0.311	0.323	0.342	0.346	0.360	(40)	
		taud	2.521	2.529	2.575	2.598	2.616	2.619	2.616	2.593	2.538	2.504	2.518	2.541	(41)	
	Ebn at Noon		958	942	936	914	899	887	905	941	966	977	990	984	(42)	
		Edh at Noon	113	110	101	92	83	79	82	90	103	112	113	111	(43)	
(44) All-Sky Solar Radiation	RadAvg		7.90	7.48	6.58	5.47	4.55	4.00	4.27	5.37	6.57	7.54	8.20	8.30	(44)	
	RadStd		0.34	0.37	0.39	0.31	0.28	0.24	0.35	0.20	0.19	0.21	0.20	0.36	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	+0.31	N/A	-1.02	N/A	N/A	N/A	N/A	N/A	+153	+160	(46)	
(47)	Regional (3 neighbors)	+0.31	N/A	-1.02	N/A	N/A	N/A	N/A	N/A	+153	+160	(47)	

Nomenclature: See separate page